

Mahault Albarracin: “Sustainability Under Active Inference”

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all right the next video is going to be 28 minutes long it's from ma alasen sustainability under active inference hey everyone my name is maras PhD candidate at the University Quebec and director of research strategy at versus today I'm going to be talking to you about budding work on sustainability under active inference which we had started with multiple authors that I listed down below uh starting from the core components and scaling up to a computational model we are pursuing this research in multiple collaborations ADV verses around sustainability ultimately this is meant to Charter a path to much larger projects so feel free to reach out if if you want to collaborate so let's begin with the most fundamental aspect of sustainability namely resilience resilience refers to an agent or system's ability to persist and adapt in the face of challenges or perturbations under active inference resilience is a multifaceted concept involving different strategies that allow a system to maintain or recover functional stability resilience can be broken down into three core components inertia elasticity and plasticity so let's start by defining inertia it describes an agent's capacity to resist change when subjected to external disturbances it really is the stability and robustness of a system in active inference inertia is represented by high Precision beliefs which can be thought of as highly confident low entropy beliefs about the system's current state inertia is reflected in the Precision weighing of prediction errors higher Precision implies stronger resistance to updating beliefs maintaining a stable State uh means that Precision is implemented as a scaling Factor so here higher Precision really implies increased resistance to change the problem is that inertia alone makes systems brittle errors they cannot adapt so then we move to the second component elasticity elasticity describes the system's ability to return to characteristic or retractor States after a perturbation it corresponds to the concept of homeostasis so the capacity to adaptively recover in active inference elasticity is expressed in terms of an agent's ability to minimize free energy and seek stable States it can be mapped onto the relaxation back to attractor states which is driven by minimizing free energy as the system follows a trajectory in doing so it

restores the system to a preferred state analogous to returning to an attractor point so for example in nature biological systems really exhibit elasticity through homeostatic feedback mechanisms like temperature regulation or blood glucose balance these systems have the ability to change but they ultimately return to where uh they were without necessarily learning and this may yield the same errors and ultimately can crap Under Pressure much like a rubber band pulled too hard so let's move on to the third plasticity it refers to the expansion of an agent's repertoire of adaptive States it allows systems to adapt not only by returning to previous States but also by exploring new States enhancing robustness so in active inference plasticity is related to functional redundancy and structural degeneracy having multiple Pathways to achieve the same outcome structural degeneracy means that you have different components perform overlapping roles allowing the system to adapt to novel challenges but we will dig into this a little later so here plasticity is captured through posterior beliefs that can change flexibly based on new sensory input or structure them if you actually add a new state we can use the variational density such that the agent can represent a diverse range of potential States maintaining a dynamic model of its environment think for example of neural plasticity where different neural circuits can adapt to perform similar function after damage to understand how plasticity comes into play we really have to hone in on certain key concept underlying it so complexity refers to the uncertainty or diversity of System Dynamics in active inference complexity is associated with the range of possible States an agent must consider complexity is linked to entropy of the system State distribution higher entropy reflects greater uncertainty and the number of potential States the agent must account for variational free energy is a way of balancing two important things how complex our beliefs are and how accurately they match what we observe and so we can balance this in a formula that goes f equals the difference between the complexity of updating our beliefs and how accurately these beliefs predict what we observe so this formula the first part DK measures the complexity or the cost of updating our beliefs to align with reality the second part written as EQ represents how well our current beliefs can predict the observations we make so we said we were going to turn to degeneracy and redundancy and I'm obviously here leveraging work um by no uh and Karl Friston for degeneracy what we're considering is that the presence of different components are capable of fulfilling similar roles within a system in her paper they talked about for example left hand right hand and redundancy means that you have similar components performing the same function which enhances robustness against failure the redundancy here involves overlapping functions in state transition matrices B which indicates

similar transitions across different states and degeneracy can be Quantified by the entropy of the distribution which represents the number of alternative configurations that lead to similar outcomes and so managing complexity through diverse beliefs really means that the agents can adapt to changing conditions degeneracy means the agent can switch strategy when one pathway becomes unavailable which promotes adaptation and redundancy ensures the continuity function when certain components fail which adds resilience to the system so degeneracy here helps reduce the complexity cost by Distributing probabilities across multiple States thus maintaining low free energy and supporting robustness in diverse conditions but while resilience is really critical for maintaining stability it doesn't fully capture the dynamic interaction between systems and their environments resilience focuses on an agent's ability to withstand or recover from perturbations but it lacks insight into the broader System Dynamics without considering systemic interdependencies focusing solely on resilience can lead to shortterm adapt ation without promoting long-term stability this is work that Mark Miller and inisha Palo have uh written a lot about so for sustainability we really need that um systems not only adapt but also ensure the long-term viability of their environment and resource base we can through active inference integrate resilience and sustainability and and show the need for systemwide coordination for harmonious interaction as Carl likes to say active inference really is just the story of resilient and sustainable systems so let's Define sustainability a little more sustainability is defined as the enduring capacity to meet needs without depleting crucial resources and here resources is taken a little more liberally we can expand Beyond material resources and Encompass social ecological and System Dynamics sustainable systems are those that persist over time and do so in a manner that promotes stability and health across all dimensions environmental economic and social we achieve this under active inference by enhancing longterm adaptability and interconnectedness across multiple scales sustainability emphasizes the creation of conditions that allow for continued adaptation without exhausting resources effectively ensuring that all parts of the system work cohesively to support resource renewal and plasticity so sustainability can be framed as optimizing the expected free energy over deep temporal scales which is work that Casper hesp uh has pioneered balancing immediate needs with long-term system viability so the optimization of expected free energy can be represented as um f of expected uh SMA equals the risk plus the ambiguity of X of f expected s so risk quantifies the Divergence between the predicted paths I.E the future states of the system given an action sequence and past observations and and the preferred paths which are ideally sustainable and align with long-term goals this encourages the system to

pursue states that meet its sustainability preferences over time ambiguity measures the expected uncertainty in mapping states to observations and it ensures that the system continuously learns about its own Dynamics and the environment this helps the system reduce UNC and improve its predictive Dynamics um over time the components of expected free energy balance the drive to fulfill preferences with the need to minimize uncertainty and so this balance supports plastic responses that enable the system to adapt to change but also actively guide itself towards more sustainable and self-organizing configurations over extended time scales ensuring resilience at both individual and Collective levels but sustainability isn't just about survival or at least it's not a simple story it must incorporate UD dionic well-being focused on growth fulfillment and development and honic well-being which captures immediate satisfaction so in active inference this means that the system should balance present experiences with long strm strategies that enhance overall capacity for adaptive growth so when we talk about a dionic well-being we represent growth and the system uh the system's ability to find diverse beneficial States and it aligns with long-term sustainability by ensuring that in survive and thrive through extended adaptability and resource renewal it represents the growth and systems ability to achieve it's long-term goals this this concept aligns with active inference because it emphasizes long-term sustainability uh by optimizing internal models that allow agents to find those beneficial States and So within active inference agents act to minimize expected free energy which involves the balance as we saw between risk minimization which means that the outcomes are aligned with the agent's preferences and ambiguity minimization which ensures that the learning continuously happens so that's what enables the agent to grow and Thrive and so UD dionic well-being can be conceptualized as the successful maintenance of favorable States and the expansion of the system's potential allowing it to reach higher order goals over extended time Horizons and so this Behavior reflects structure learning wherein agents adapt to changes in the environment by restructuring their inter internal models which lead to growth in a diverse set of beneficial States now honic well-being involves immediate momentto moment satisfaction which is necessary to maintain ongoing motivation and effective response to environmental demands it focuses on immediate satisfaction and so the agent's capacity to respond adaptively to current needs uh it can be seen in active inference as the system's ability to reduce momentary free energy and maximize instantaneous rewards so we can see this at the lower level say of hierarchical models so you minimize prediction errors in the present you seek sensory states that satisfy immediate needs or preferences like finding food to satisfy hunger and this

corresponds to reducing free energy associated with specific observations the internal reward or momentary satisfaction that the agent achieves after fulfilling immediate needs corresponds to the agent reaching a preferred State as predicted by its internal model this is crucial for maintaining motivation and ensuring effective responses to environmental demand here we also have to factor interdependencies within the system these interdependencies can be represented through mutual information equations which ultimately quantify the connections between different components the plasticity at different individual levels translate to the sustainability of the entire system system aligning short-term actions with long-term goals so sustainability is an emergent property that depends on the balance between all elements in a system this really involves preventing resource depletion maintaining well-being and promoting resilience through self-organization and learning so it's a continuous effort to increase resource redundancy and avoid over-exploitation thereby ensuring sustainable practices over time you can think of it as avoiding a local minimum the sustainability of a system depends on harmonious interactions among its different components which range from individual agents to the broader collective this means that the system must align both internal and external goals to ensure stability over extended periods enhancing resilience and sustainability so for example the resilience components of inertia elasticity and plasticity will work synergistically to support adaptive actions that contribute to sustainable outcomes the emergence of stability is promoted through decentralized interactions where each component of the system self-organizes and responds to environmental feedback which reduces the complexity and increases adaptive capacity and allows the system to maintain a dynamic but stable equilibrium also sometimes referred to as homeostasis so we can think of attractor states which emerge as agents minimize their variational free energy over time and this process supports the alignment of internal models with the environmental demands ensuring that agents remain adaptive while reinforcing the renewal of resources and overall system well-being so these are all nice ideas and you know we think they hold up mathematically but we wanted to test them out and obviously this is the beginning of the journey so we built a small and very simple agent based model to illustrate sustainability under active inference at the level of one single agent within one environment this is extremely simple and the following studies will cover more so the model here simulates an agent managing food consumption in different environments to ensure long-term resource availability our objective was to show that the agent's goal is to maximize well-being while ensuring resource sustainability over time and this involves balancing immediate needs with

uh future availability of resources so we started with a validation case um in this case the agent makes decisions uh based on current food availability without affecting future availability and so the agent observes food and decides whether to consume or not uh based on its current state and prior preferences so we have the generative model uh we have hidden States representing food availability and saity and the preference uh C which encoded a strong preference for saity while maintaining food availability as as you can see these two preferences are technically contradicted and so in a static environment which was the first case the agent decisions are really straightforward they just maximize short-term well-being without needing to manage resource Dynamics so as you can see uh in the tiny little plots here uh the we can see that the agent's Behavior specifically uh whether the agent chooses to eat or not over a series of time steps and so uh the middle plot shows that the amount of food over time which is represented as food left and the last bottom plot shows that uh the uh you represent the agent's level of SA or how satisfied or full the agent feels over time higher values indicate higher levels of sa and so uh the agent uh has a strong preference for high sty uh since food is continuously available the agent keeps eating at the beginning which increases its saity level and since the environment is static the food remains available indefinitely and so the agent successfully increases its H uh we then wanted to show whether or not what we did was just an artifact of the Simplicity of the model so we screwed with its matrices and so incorrect matrices cause the agent to misjudge the effect of eating which leads to an unstable sa level so we we showed how errors in perception and action models can significantly impact the ability of an agent to achieve stable and desirable States even in an extremely simple uh setup so then we tried something a little more complicated which is a dynamic environment where the strategies have to change so uh the environment changes based on the agent's actions so resources deplete when consumed but replenish over time when not used and so the agent's challenge was that it had to learn to balance consumption with replenishments uh which effectively had to demonstrate some degree of sustainable Behavior so the model includes the transition Matrix that Maps the state of food available based on the agent's actions and expected free energy here minimization uh guides the agent to consider future State and ensure availability um so this was the case as you can see it hasn't changed much the only thing that has changed is the Dynamics of the environment and the ability for the agent to ultimately learn those Dynamics so the agent had to ad adaptive strategies that balanced immediate rewards with long-term stability uh and this was showing us how active inference can foster sustainable behaviors in fluctuating environments um so when we look at this plot um we

could see that the agent uh has learning turned on or not and we wanted to see whether an agent that didn't learn did just as well as an agent who learn just to see again whether it was an artifact of the Simplicity of our model and also the agent can die if the resources are depleted and it has high sa it is not satiated over time it dies which you know ethically might not have been the best thing to do but whatever so agents that have learning quickly improve their survival Times by refining their internal models and strategies and so the rapid Improvement really shows that learning enables them to find efficient and sustainable ways to interact with their environment without learning there is no significant increase uh in survival time uh as they unable to really adapt their behavior based on past experiences so they end up uh in suboptimal decision Loops leading to a lower average survival so we can really show that learning is crucial for agents's ability to adapt which in turn allows for a longer more sustainable uh presence in the environment and ultimately uh you know the also the environment's ability to actually be sustainable in itself the agent overcomes initial limitations eventually converges to strategies that maximize our survival which really shows the power of dynamic adaptation in uncertain environments so we saw that the agent's behavior in both static and dynamic environments really shows how well-being can be optimized while maintaining resource availability and in the dynamic environment the agent learns to moderate consumption it is able to understand the tradeoff between immediate satisfaction and future resource sustainability the agent has the ability to create adaptive strategies that adapt uh that align sorry with the principles of resilience and sustainability and so here our model illustrates in a very very minimal way how minimizing free energy can lead to behaviors that promote long-term viability rather than short-term gains and so these simple models are just laying the groundwork for modeling more complex systems such as ecosystems or Urban environments where sustainability requires coordinated interactions between multiple agents and resources so to recap we started by understanding resilience with an active inference we broke it down to inertia elasticity and plasticity we then explored sustainability building on resilience and we incorporated well-being as a component we demonstrated how agents can balance immediate needs and future viability achieving sustainability again all through active inflece and so it Maps systems that can adapt sustain and Thrive by balancing short-term adaptation with long-term sustainability but we still need to show how active inference can effectively be scaled to real world applications involving multiple interacting agents and resources uh involving the ability for the environment to fully deplete and therefore how to actually manage such possible um such possible patterns and how agents might learn over different areas um we have to

transition from theoretical models to more practical implications that involve challenges in data acquisition model complexity computational scalability um we also need to factor Mutual interactions and the effects of each subsystem on each other and the larger system such that they promote more resilience and plasticity so all of this is extremely seminal we have to move uh further and actually demonstrate in much larger scale and more complex simulations this these ideas but it has critical potential applications achieving all this we can use active inference to model and predict adaptive strategies for mitigating climate impacts at different scales or improve decision-making processes in organization uh enhancing resilience and sustainability in economic and social contexts so active inference can be extended to other complex adaptive system and we can support initiatives aimed at fostering sustainability in health care Agriculture and urban planning which is work that again Mark Miller and Ben White tenesi poo uh are really pushing forward so thank you for listening and I hope you enjoyed